Resting-state fMRI noise regression in multi-site aging studies

Norman Scheel[1], Jeffrey N. Keller[4], Ellen F. Binder[5], Eric D. Vidoni[3], Jeffrey Burns[3], Binu Thomas[2], Diana R. Kerwin[6], Wanpen Vongpatanasin[2], C. Munro Cullum[2], Rong Zhang[2,6], David C. Zhu[1]

[1]Michigan State University, East Lansing, MI, USA,
[2]UT Southwestern Medical Center, Dallas, TX, USA,
[3]University of Kansas Alzheimer's Disease Center, Fairway, KS, USA,
[4]Pennington Biomedical Research Center, Baton Rouge, LA, USA,
[5]Washington University School of Medicine, St. Louis, MO, USA,
[6]Texas Health Presbyterian Hospital, Dallas, TX, USA

Introduction

Risk reduction for Alzheimer's Disease (rrAD) is an on-going trial designed to assess the effects of aerobic exercise and pharmacological cardio-vascular control on cognitive function in older adults who are at a higher risk of developing Alzheimer's Disease (Szabo-Reed et al. 2019). As part of the neuroimaging protocol, resting-state fMRI (rs-fMRI) is obtained at baseline and again after two years of intervention. Our aim was to find the most efficient rs-fMRI noise regression technique across scanners and rrAD subjects, as noise regression techniques are still diverse and much debated. Also, emerging methods like AROMA (Pruim et al. 2015a) and SOCK (Bhaganagarapu et al. 2013) have only been evaluated on young subjects (Pruim et al. 2015b). Here, we compared the performance of six common techniques using metrics defined as in Pruim et al. 2015b and Bhaganagarapu et al. 2013.

MRI system	head coil channels	voxel size in mm	number of subjects	age (years, mean±SD)	motion RMS-FD (mm, mean±SD)
Philips Achieva	32	3.4 x 3.4 x 3	49	69.1±6	0.116±0.072
GE MR750W	24	3.4 x 3.4 x 3.4	61	69.4±6	0.107±0.077
GE MR750W	32	3.4 x 3.4 x 3	128	68.5±6	0.096±0.055
Siemens Skyra	32	3.4 x 3.4 x 3	106	68.2±5	0.108±0.07
Siemens Prisma	32	3.4 x 3.4 x 3	90	68.6±6	0.116±0.062

SD: standard deviation, RMS: root mean square, FD: frame-wise displacement

- 5 3T MRI systems
- 284 volumes (12min)
- 2.5s TR
- 28ms TE
- 64x64 matrix
- 434 subjects
- age: 68.6±6 years
- frame-wise displacement: 0.116±0.072 mm

fMRI noise regression techniques for comparison

A
spike regression
AFNI's "3dDeconvolve" regressors:
- 24 motion parameters (24MP)
- Spike regr ***FD<0.5mm***
- white-matter (WM) and cere-bro-spinal fluid (CSF) signals
- bandpass filter (0.009-0.08Hz)
- third order detrend

B
spike regression + GSR
AFNI's "3dDeconvolve" regressors:
- 24 motion parameters (24MP)
- Spike regr FD<0.5mm
- WM and CSF signals
- ***global signal***
- bandpass filter (0.009-0.08Hz)
- third order detrend

C / D
aggressive / non-aggressive AROMA
AROMA
- ***aggressive*** and ***non-aggressive***

AFNI's "3dDeconvolve" regressors:
- bandpass filter (0.009-0.08Hz)
- third order detrend

E
one-step AROMA
AFNI's "3dDeconvolve" regressors:
- ***AROMA regressors***
- bandpass filter (0.009-0.08Hz)
- third order detrend

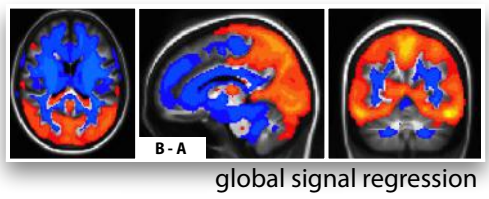
F
SOCK
AFNI's "3dDeconvolve" regressors:
- ***SOCK regressors***
- 24 motion parameters (24MP)
- WM and CSF signals
- bandpass filter (0.009-0.08Hz)
- third order detrend

Results

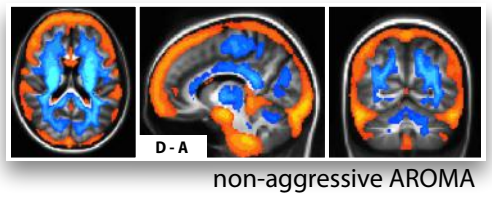
Resting state network (RSN) reproducibility, assessed by 500 split-half permutation tests, has the highest score in aggressive AROMA, RSN identifiability (true positive) has best score in SOCK and for false positive in SOCK and one-step AROMA. Edge activity scores best in SOCK and spatial smoothness fractions score best in aggressive AROMA. Motion scrubbing regression (A,B) had a loss of tDOF of 7±4%, AROMA (C,D,E) lost 15±4% and SOCK (F) lost 10±2%. Global mean regression shows unbalanced anterior to posterior changes in tSNR. On the contrary, aggressive AROMA enhances tSNR in grey matter and reduces tSNR in WM and CSF regions uniformly across the brain. The performance of all techniques on the data of each scanner separately was comparable with all subjects' data combined.

	RSN Reproducibility *z-score*	RSN Identifiability (true positives) *z-score*	RSN Identifiability (false positives) *z-score*	Edge activity %	CSF activity %	Spatial smoothness	
A spike regression	5.8±1.2	3.5±1.1	0.4±0.2	11±9	17±5	0.028±0.004	best
B global mean regression	5.3±0.7	3.4±1	0.5±0.2	8±4	15±4	0.023±0.004	
C aggressive AROMA	**6.9±1.2**	3.6±1	0.6±0.2	10±5	20±6	**0.031±0.007**	
D non-aggressive AROMA	4.5±1.4	1.6±1.2	0.5±0.2	26±19	25±11	0.03±0.007	
E one-step AROMA	6.4±1.3	3.5+1.2	**0.4±0.1**	6±4	**12±4**	0.022±0.006	
F SOCK	6.4±1.3	**4.1±1**	**0.4±0.1**	**6±2**	13±3	0.023±0.003	worst

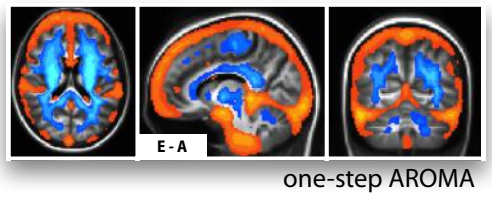
Temporal signal to noise ratios in comparison to technique A



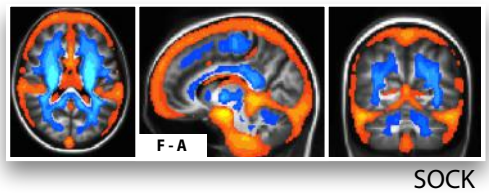
global signal regression



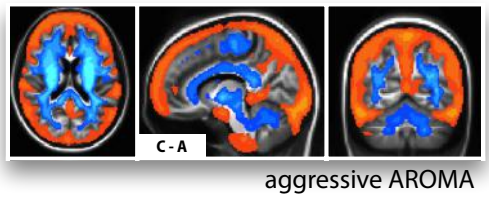
non-aggressive AROMA



one-step AROMA



SOCK



aggressive AROMA

Conclusions

Comparing across all data quality metrics, most techniques performed satisfyingly with marginal differences, except technique D (non-aggressive AROMA), which performed worst. SOCK performs best in RSN identifiability and edge activity, yet considering reproducibility as the most important factor for longitudinal studies, technique C (aggressive AROMA) appears to be the most favorable noise regression technique across scanners and for older hypertensive subject populations as in the rrAD trial.

References

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